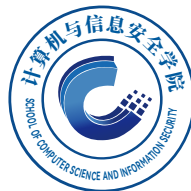


双周工作汇报

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① 汇报简介与目标论文背景

② 问题回答

③ 图片总览

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Introduction

经过第二次汇报，在结构优化的前提下，对文章内容的解读仍有缺陷。本次着重讨论如下问题：

- 文章聚焦的核心工程问题是什么？
- 所构建的多模态大模型，其核心决策任务是什么？输出什么？

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Q1

本文要解决的核心工程问题是：基于多源数据（图像、信号、文本等），为工业场景中的设备维护、故障诊断、缺陷检测等任务，提供智能且高效的决策支持方案。实例如下图所示

Q1



Time stamp	Product ID	Cycle Time (s)	Inspection Result	Size (Dimension 1 (mm))	Size (Dimension 2 (mm))
2023-10-01 08:00	00001	120.5	Pass	20.0	10.0
2023-10-01 08:05	00002	118.9	Pass	20.1	9.9
2023-10-01 08:10	00003	121.0	Pass	19.8	10.1
2023-10-01 08:15	00004	119.5	Pass	20.2	9.8
2023-10-01 08:20	00005	122.1	Pass	19.9	10.2
2023-10-01 08:25	00006	120.8	Pass	20.3	9.7
2023-10-01 08:30	00007	119.2	Pass	19.7	10.3
2023-10-01 08:35	00008	121.5	Pass	20.4	9.6
2023-10-01 08:40	00009	120.0	Pass	19.6	10.4
2023-10-01 08:45	00010	122.5	Fail	20.5	9.5
2023-10-01 08:50	00011	120.3	Pass	19.5	10.5
2023-10-01 08:55	00012	119.8	Pass	20.6	9.4
2023-10-01 09:00	00013	121.2	Pass	19.4	10.6
2023-10-01 09:05	00014	120.7	Pass	20.7	9.3
2023-10-01 09:10	00015	119.0	Pass	19.3	10.7
2023-10-01 09:15	00016	121.8	Pass	20.8	9.2
2023-10-01 09:20	00017	120.4	Pass	19.2	10.8
2023-10-01 09:25	00018	122.0	Fail	20.9	9.1

The automated assembly process of the constant velocity joint is shown in <image>, and the production records are provided in <csv file>. Could you help analyze the production cycle time and quality control situation, and also extract the key parameters?



The situation on-site shows that the production cycle time is normal, but there are quality issues with the parts. In particular, Dimension 1 has sudden variations, while Dimension 2 still follows a Gaussian distribution. This suggests that the problem may be concentrated in the process or equipment affecting Dimension 1. Could you please check the relevant processes?



According to the investigation, Dimension 1 represents the pitch circle diameter of the star-shaped raceway. Could you please analyze this based on the statistical data and provide some optimization suggestions?



Since Dimension 1 (the pitch circle diameter) shows a sudden variation while Dimension 2 still follows a Gaussian distribution, it indicates that the issue with Dimension 1 is likely caused by a sudden anomaly in a single process or piece of equipment, rather than a systemic problem. I suggest comparing the raw material batches of the defective parts to check for any batch differences, and also inspecting the wear conditions of molds and fixtures.

图 1: 用于智能制造的多模式大型语言模型系统的对话示例

Q2

模型需决策的是：对输入的图像、信号、文本等工业数据进行分析与理解，判断设备状态、故障类型或缺陷情况；输出为包含分析结果、诊断结论甚至优化建议的文本或可视化内容（如标注后的图片）。系统整体框架如下图所示

Q2

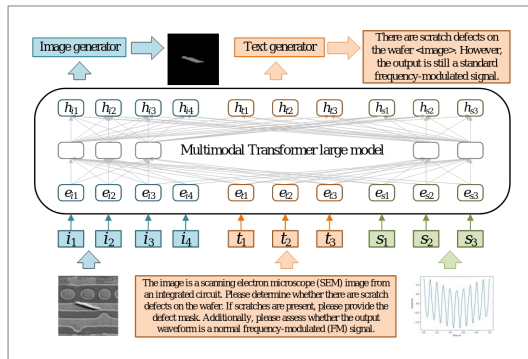


图 2: 基于多模态 Transformer 架构的统一双向图像文本生成框架

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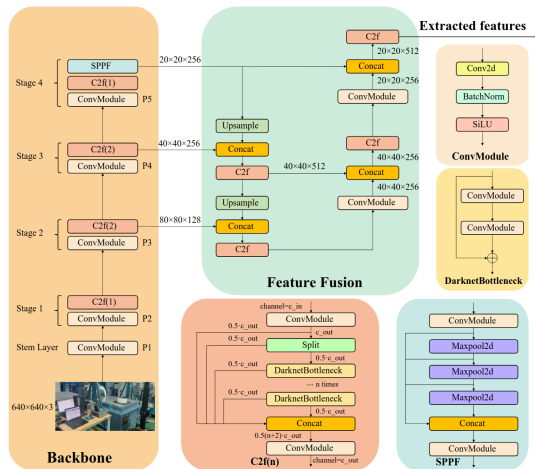


图 2: 基于 CNN 的视觉编码系统结构的全面说明

图片总览

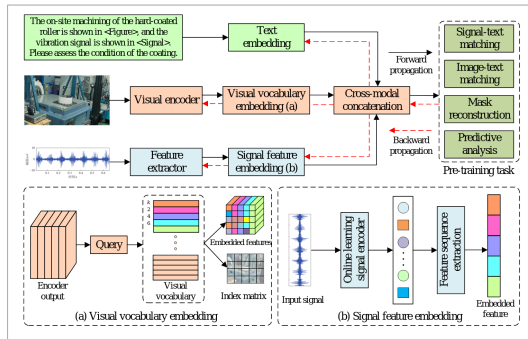


图 3: 用生成动态视觉和信号字典以弥合表征差距的工作流程。

图片总览

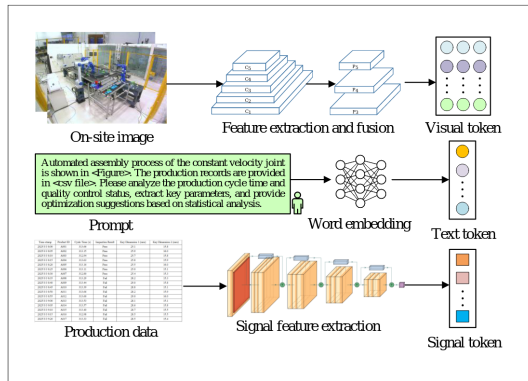


图 3: 调整多种输入方式以输入到基于 Transformer 的语言模型

WELCOME FEEDBACK!

